Mach-Zehnder Interferometer

Md. Rakibul Islam

Dept. of Electrical and Electronic Engineering

Rajshahi University of Engineering and Technology

Rajshahi, Bangladesh

Rakibul.islam@eee.ruet.ac.bd

Abstract

I. Introduction

In modern telecommunication networks, bandwidth demand is increasing exponentially and to maintain this demand integrated optical components needs to provide high-speed performance with lowering footprint and power consumption[1]. Simultaneously, the grinder of silicon photonics, electro-optic modulators and wavelength filters, must achieve low insertion loss, and minimal driving voltages.

Traditional Mach-Zehnder Interferometers (MZIs) and optical ring resonators (ORRs) are utilized for modulation and filtering and both have some fundamental limitations. A π phase shift is required for full on-off switching and to achieve the phase shift a conventional MZI requires either a highly efficient phase shifter, a long device length or a large RF voltage swing, but both result in footprint and energy penalties[1]. Besides, flat-top spectral response can not be achieved without cascading multiple stages of MZIs[2]. On the other hand, due to resonance enhancement ring resonators provides compact footprints with low-voltage operation. But high Q-factor dependencies of MRR limits the modulation bandwidth[1], [3]. Moreover, unwanted chirp gets added to the optical signal while modulating in a standard ring resonator or racetrack modulator (RTM)[4].

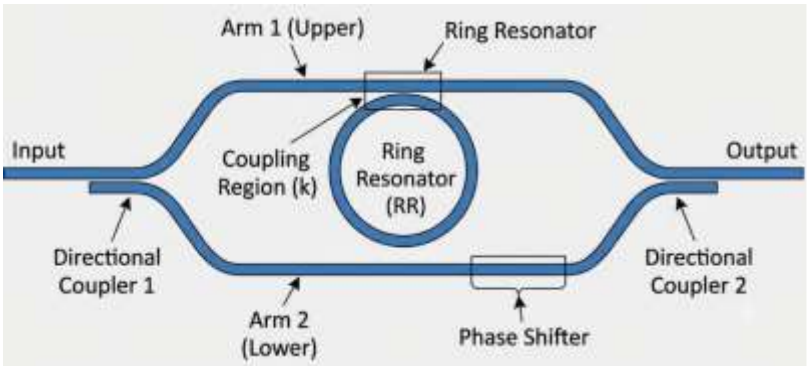


Fig. 1: The Ring-Assisted Mach-Zehnder Interferometer (RAMZI)

Integrating micro-ring resonators within the interferometric arms of an MZI, called the Ring-Assisted Mach-Zehnder Interferometer (RAMZI), overcomes the constraint smoothly[2], [3]. To compensate the MZI's intrinsic nonlinear transfer function, the Ring-Assisted Mach-Zehnder Interferometer (RAMZI) utilizes the steep phase tuning of the ring resonator[3]. Furthermore, it removes the severe bandwidth bottlenecks of the standard ring modulators and delivers chirp-free modulation[4]. In filtering applications, the RAMZI produces sharp, asymmetric Fano resonances and flat-top passbands, enabling superior wavelength selectivity[2].

I. Theory

An incoming optical electric field Einsplits into two separate waveguide arms in a standard Mach–Zehnder Interferometer (MZI). Before recombination a phase difference is introduced between the two optical paths by creating a difference in arm length, refractive index, or applied voltage. The interference between these two propagated signals result in output field [5]. The output electric field of an asymmetric MZI can be expressed as:

$$E_{out} = \frac{E_{in}}{2}[e^{-j(\omega_0 t+\varphi_1)}e^{-\alpha L_1} + e^{-j(\omega_0 t+\varphi_2)}e^{-\beta L_2}]$$

where:

- ω_0= angular frequency of the optical carrier
- $\alpha$and β= propagation loss coefficients of the upper and lower arms
- $L_1$and L_2= lengths of the interferometer arms
- $\varphi_1$and φ_2= induced optical phase shifts in each arm

The operational behavior of the MZI is governed by the phase difference:

$$\Delta\varphi = \varphi_2 - \varphi_1$$

Using an external voltage, enabling optical modulation and interference-based signal processing, effective refractive index Δn_{eff} can be modified to directly control the phase difference[5].

A single ring resonator coupled to a straight bus waveguide is commonly modeled using transfer matrix formalism. For a lossless coupler and in the coupling region,

$$\kappa = j\sqrt{1 - t^2}$$

where:

- t= transmission coefficient
- κ= cross-coupling coefficient[1]

Inside the ring the optical field experiences both attenuation and phase accumulation during propagation. The propagation factor is written as:

$$p = e^{(-\alpha + j\beta)L}$$

where:

- α= attenuation coefficient
- β= propagation constant
- L= circumference or cavity length of the ring resonator[1]

The transfer function at the through-port is given by:

$$\frac{E_T}{E_{in}} = \frac{t - p}{1 - pt}$$

The effective phase response changes rapidly at the resonance condition. The effective phase shift is defined as:

$$\phi_{eff} \equiv \arg\left(\frac{E_T}{E_{in}}\right)$$

This sharp phase variation provides extremely high phase modulation sensitivity, making ring resonators highly responsive to small refractive index changes[1].

A Ring-Assisted Mach–Zehnder Interferometer (RAMZI) accumulates the broadband interference behavior of an MZI with the highly dispersive phase response of a ring resonator[2]. Adding the ring resonator into one or both arms of the interferometer enhance spectral selectivity and phase sensitivity. If a ring resonator with complex amplitude response $H_R(\gamma,\ \theta)$is coupled into one arm of the MZI, the resulting transfer function at a particular output port becomes:

$$H_A = \frac{1}{2}\left[H_R(\gamma,\theta)e^{j\varphi_r} + e^{j\varphi_a}\right]$$

where:

- $H_R(\gamma,\theta)$= complex transfer response of the ring resonator
- φ_r= phase shift introduced in the ring-assisted arm
- φ_a= phase shift of the reference arm[6]

The ring produces sharp resonant phase response which interferes with the broadband optical response of the reference arm. The abrupt resonant phase discontinuity in the transmission spectrum converted into an asymmetric Fano-type resonance by this interaction. As a result, RAMZI structures exhibit enhanced filtering characteristics, improved modulation efficiency, and superior spectral shaping performance compared to conventional MZI configurations[7].

II. Waveguide Modal Analysis

A. Waveguide Cross-Section

Standard industry level dimension 220 nm×500 nm SOI strip waveguide is used throughout the work. The foundry process set 220 nm device-layer height and for single-mode operation for the quasi-TE polarization 500 nm width was selected which also support the quasi-TM fundamental mode. Furthermore, at this waveguide width the SiEPIC EBeam PDK provides validated S-parameter component models for grating couplers for both polarizations[8], [9].

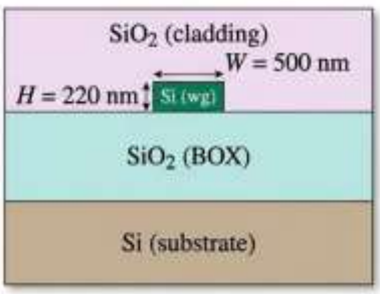


Fig. 2: Cross-section of the silicon-on-insulator strip waveguide. The silicon device-layer height is fixed by the foundry process to H = 220 nm; the design width is W = 500 nm.

A modal analysis was performed in Lumerical MODE Solutions over the wavelength range $\lambda \in [1.50, 1.60]$ µm. A 2 nm mesh resolution and perfectly matched layers (PML) boundary condition is used to suppress spurious back-reflections.

B. TE Polarization

The fundamental quasi-TE mode (intensity of the dominant E_x field component) is shown in Fig. 3.

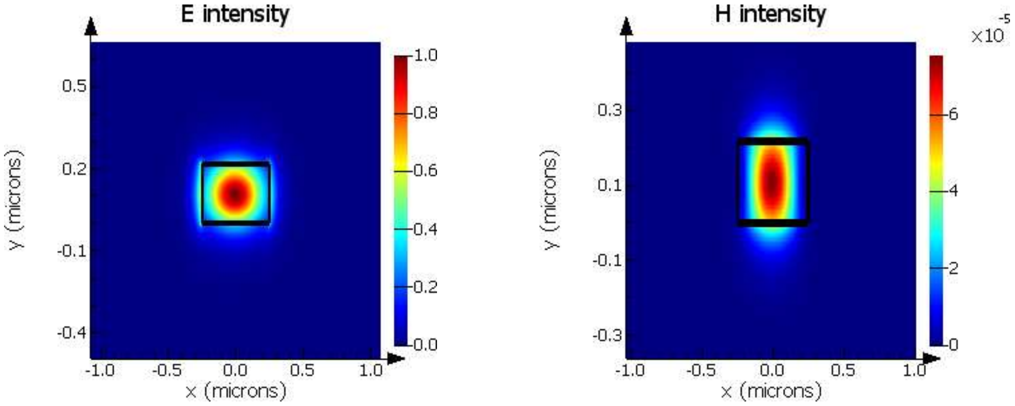
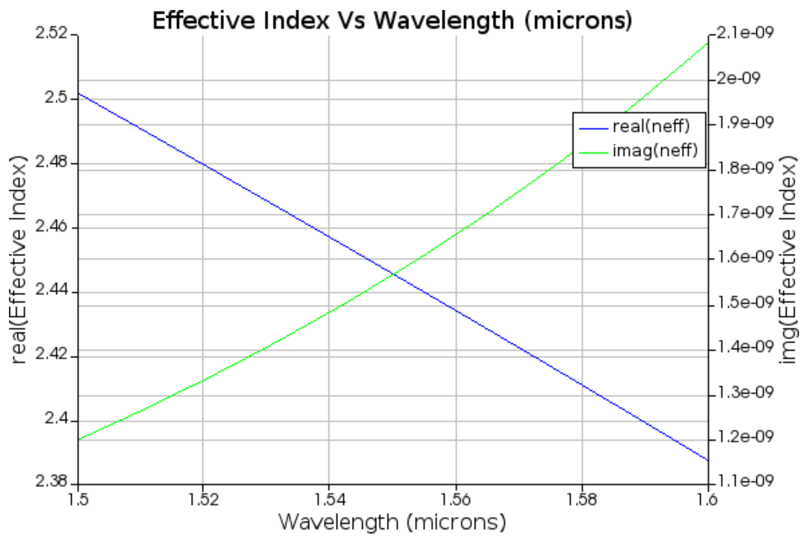
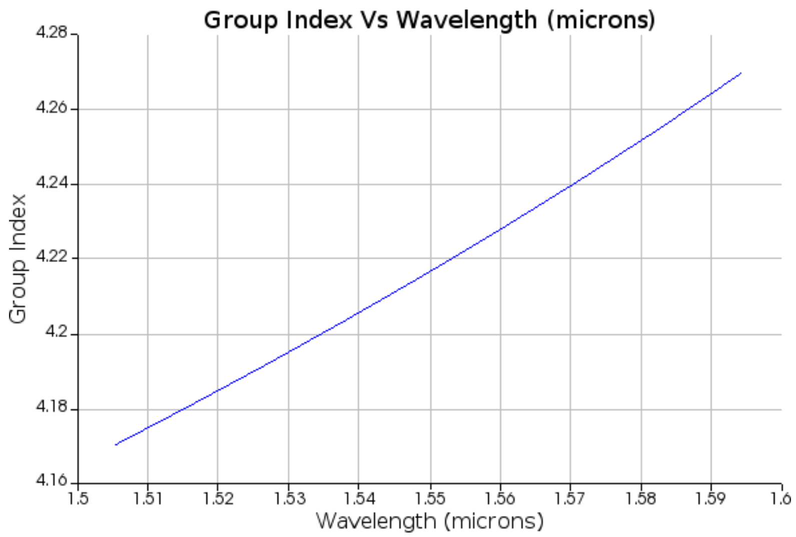


Fig. 3: Simulated E intensity & H intensity profile of the fundamental quasi-TE mode of the 220 nm×500 nm SOI strip waveguide at λ = 1550 nm. The black rectangle marks the silicon core.

The mode is well confined within the silicon core, with evanescent tails extending a few hundred nanometers into the SiO_2 cladding. Then a wavelength sweep was performed and the resulting effective and group indices are plotted in Fig. 4.



(a)



(b)

Fig. 4: (a) Effective index n_{eff} and (b) group index n_g of the fundamental quasi-TE mode versus wavelength. Both curves are extracted from a Lumerical MODE wavelength sweep on the 220 nm×500 nm SOI strip waveguide.

III. MRR Assisted MZI Circuit-Level Simulation

A. Simulation Framework

All circuit-level simulations were performed in Lumerical INTERCONNECT using SiEPIC circuit simulation tool in Klayout. Each filter comprised of one input grating coupler, one single bus waveguide, three cascaded micro ring resonator and one output grating coupler. All the components were used from SiEPIC EBeam PDK.

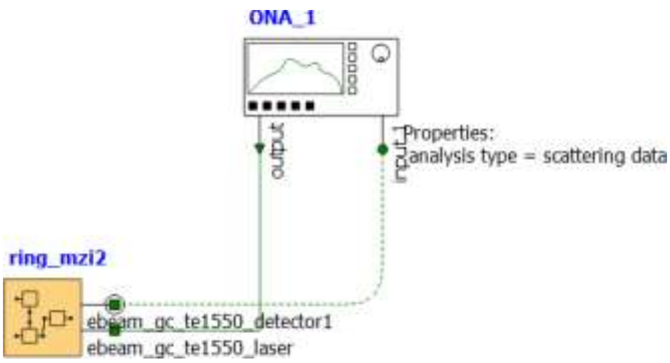


Fig. 5: Circuit layout for simulation in Lumerical INTERCONNECT

B. MRR Assisted MZI Simulation

The simulated for TE MRR assisted MZI devices have radius and gap between straight waveguide and the ring sweeps. The upper row devices are varied with radius while keeping the gap constant. The lower devices also varied with radius but in different gap constant. The spectrum of all the three devices is plotted bellow.

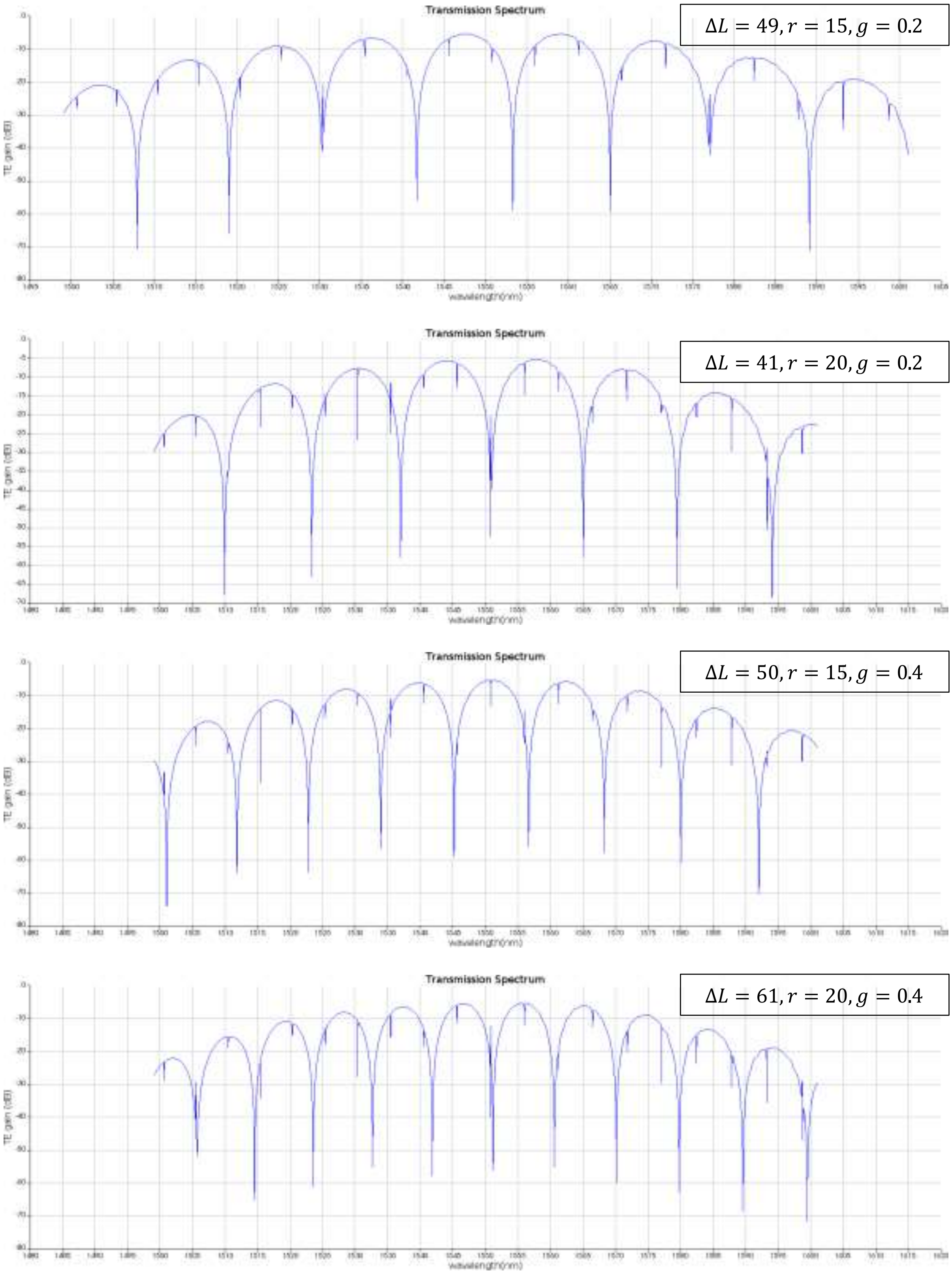


Fig. 6: Simulated transmission spectra of MRR assisted MZIs

TABLE I

TE MRR assisted MZI Designs and FSRs at λ = 1550 nm (n_g(TE) = 4.2163)

Device	ΔL (μm)	Radius, r (μm)	Gap, g (μm)	FSR (nm)

Ring_mzi2	49.298	15	0.2	11.34
Ring_mzi3	40.918	20	0.2	12.65
Ring_mzi5	50.118	15	0.4	6.39
Ring_mzi6	61.238	20	0.4	4.86

IV. Mask Layout

A. Floor-Plan Overview

The whole layout was designed in KLayout using the SiEPIC EBeam PDK. A 605µm×410µm floor consisting of 2 different types of devices. The right-side devices within yellow boarder box are MRR assisted MZI which were explored here. And a loopback structure also presents there to measure grating coupling loss.

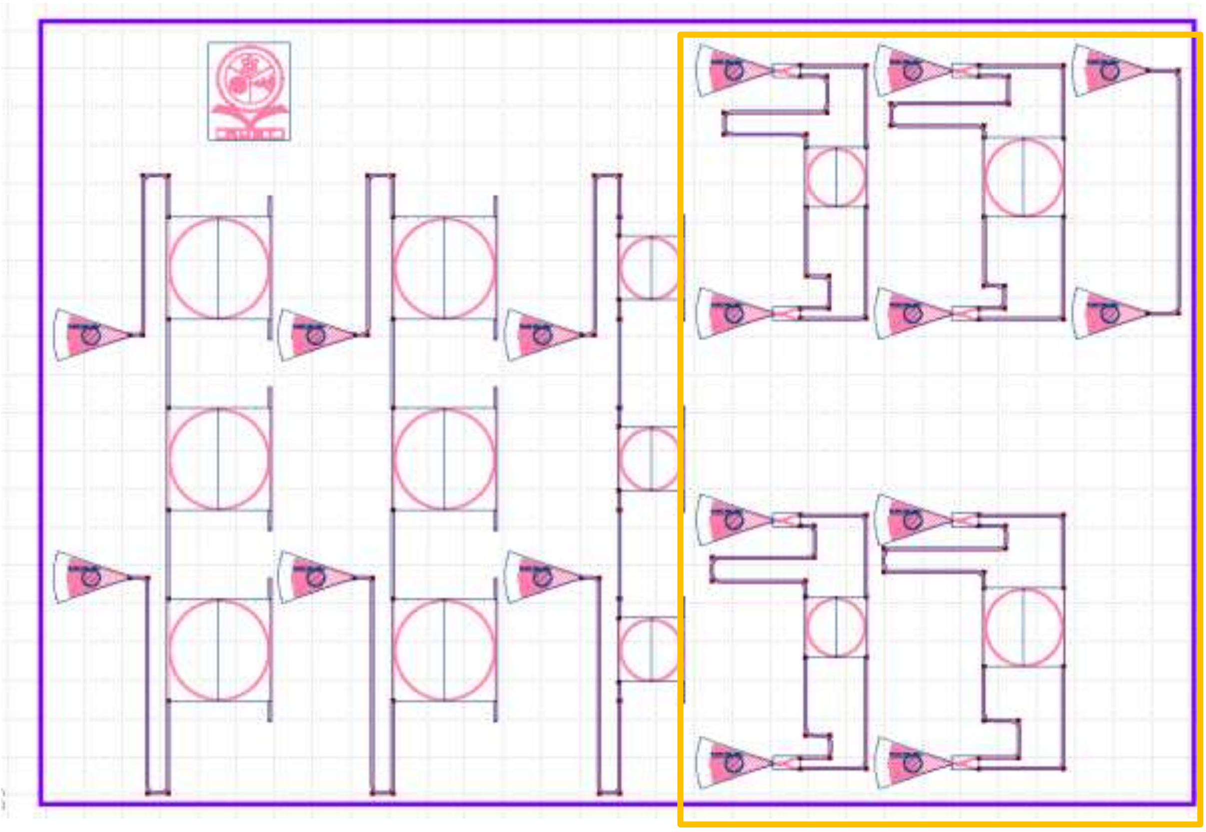


Fig. 7: Schematic top-view of the mask layout. right-side devices within yellow boarder box are MRAMZI.

- Upper row having 2 devices which are varied with radius from 15µm to 20µm at 0.2µm gap between straight waveguide and the ring for both arms.
- The lower 2 devices are also varied with radius from 15µm to 20µm but gap between straight waveguide and the ring for both arms is 0.4µm now.

The input/output pair of each cells containing TE grating couplers separated by the standard 127µm fiber-array pitch.

B. Design Rules and Conventions

The following design rules were maintained during the layout design in accordance with the SiEPIC EBeam PDK design-review checklist:

- To prevent chip rotations during automated measurement runs, all grating couplers faced in the same direction (mouth to the left)
- The vertical spacing between any two grating couplers is fixed at 127µm to match the V-groove fiber array used by the UBC measurement station.
- All the waveguide bends use circular arcs of radius well above than 5 µm as it is minimum recommended for strip TE waveguide on the SiEPIC platform. No 90° right-angle corners are present anywhere in the layout.
- The minimum feature is well above the 60 nm minimum-feature-size limit of the EBeam process.
- Every device carries a unique optical-input test label.
- To avoid optical cross-talk between adjacent waveguides a minimum 2µm distance is maintained.

V. Fabrication

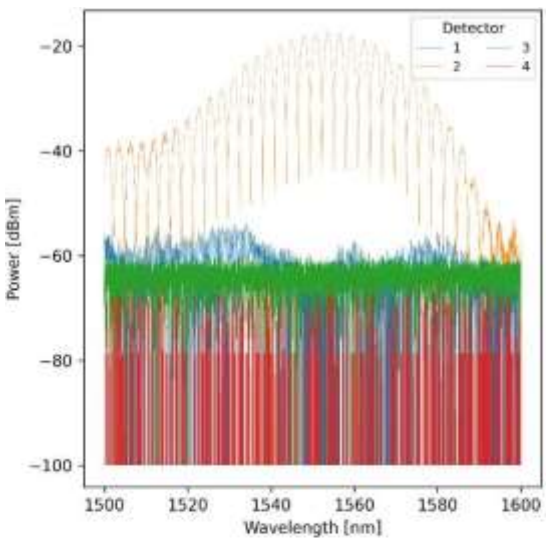
a) **Applied Nanotools, Inc. NanoSOI process:** The photonic devices were fabricated using the NanoSOI MPW fabrication process by Applied Nanotools Inc. (http://www.appliednt.com/nanosoi; Edmonton, Canada) which is based on direct-write 100 keV electron beam lithography technology. Silicon-on-insulator wafers of 200 mm diameter, 220 nm device thickness and 2 µm buffer oxide thickness are used as the base material for the fabrication. The wafer was pre-diced into square substrates with dimensions of 25x25 mm, and lines were scribed into the substrate backsides to facilitate easy separation into smaller chips once fabrication was complete. After an initial wafer clean using piranha solution (3:1 H_2SO_4:H_2O_2) for 15 minutes and water/IPA rinse, hydrogen silsesquioxane (HSQ) resist was spin-coated onto the substrate and heated to evaporate the solvent. The photonic devices were patterned using a JEOL JBX-8100FS electron beam instrument at The University of British Columbia. The exposure dosage of the design was corrected for proximity effects that result from the backscatter of electrons from exposure of nearby features. Shape writing order was optimized for efficient patterning and minimal beam drift. After the e-beam exposure and subsequent development with a tetramethylammonium sulfate (TMAH) solution, the devices were inspected optically for residues and/or defects. The chips were then mounted on a 4” handle wafer and underwent an anisotropic ICP-RIE etch process using chlorine after qualification of the etch rate. The resist was removed from the surface of the devices using a 10:1 buffer oxide wet etch, and the devices were inspected using a scanning electron microscope (SEM) to verify patterning and etch quality. A 2.2 µm oxide cladding was deposited using a plasma-enhanced chemical vapour deposition (PECVD) process based on tetraethyl orthosilicate (TEOS) at 300ºC. Reflectrometry measurements were performed throughout the process to verify the device layer, buffer oxide and cladding thicknesses before delivery.

b) **Washington Nanofabrication Facility (WNF) silicon pho-tonics process: The** devices were fabricated using 100 keV Electron Beam Lithography [1]. The fabrication used silicon-on-insulator wafer with 220 nm thick silicon on 3 µm thick silicon dioxide. The substrates were 25 mm squares diced from 150 mm wafers. After a solvent rinse and hot-plate dehydration bake, hydrogen silsesquioxane resist (HSQ, Dow-Corning XP-1541-006) was spin-coated at 4000 rpm, then hotplate baked at 80 ◦C for 4 minutes. Electron beam lithography was performed using a JEOL JBX-6300FS system operated at 100 keV energy, 8 nA beam current, and 500 µm exposure field size. The machine grid used for shape placement was 1 nm, while the beam stepping grid, the spacing between dwell points during the shape writing, was 6 nm. An exposure dose of 2800 µC/cm2 was used. The resist was developed by immersion in 25% tetramethylammonium hydroxide for 4 minutes, followed by a flowing deionized water rinse for 60 s, an isopropanol rinse for 10 s, and then blown dry with nitrogen. The silicon was removed from unexposed areas using inductively coupled plasma etching in an Oxford Plasmalab System 100, with a chlorine gas flow of 20 sccm, pressure of 12 mT, ICP power of 800 W, bias power of 40 W, and a platen temperature of 0 ◦C, resulting in a bias voltage of 185V. During etching, chips were mounted on a 100 mm silicon carrier wafer using perfluoropolyether vacuum oil. Cladding oxide was deposited using plasma enhanced chemical vapor deposition (PECVD) in an Oxford Plasmalab System 100 with a silane (SiH4) flow of 13.0 sccm, nitrous oxide (N2O) flow of 1000.0 sccm, high-purity nitrogen (N2) flow of 500.0 sccm, pressure at 1400mT, high-frequency RF power of 120W, and a platen temperature of 350C. During deposition, chips rest directly on a silicon carrier wafer and are buffered by silicon

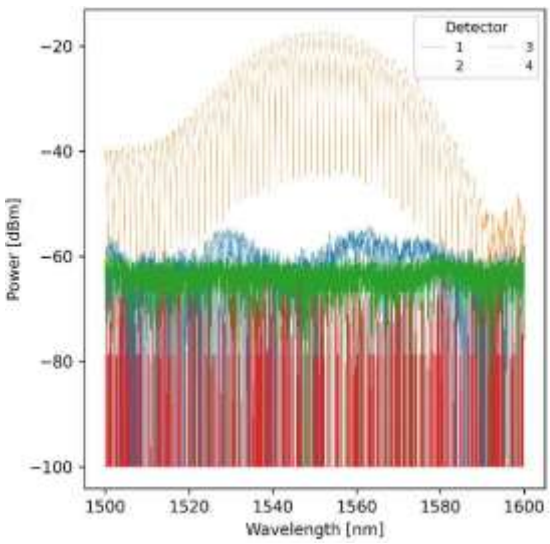
VI. Fabricated Data

The fabricated data was measured by the UBC automated measurement station and the transmission spectrum are plotted bellow.

$\Delta L = 49, r = 15, g = 0.2$



$\Delta L = 41, r = 20, g = 0.2$



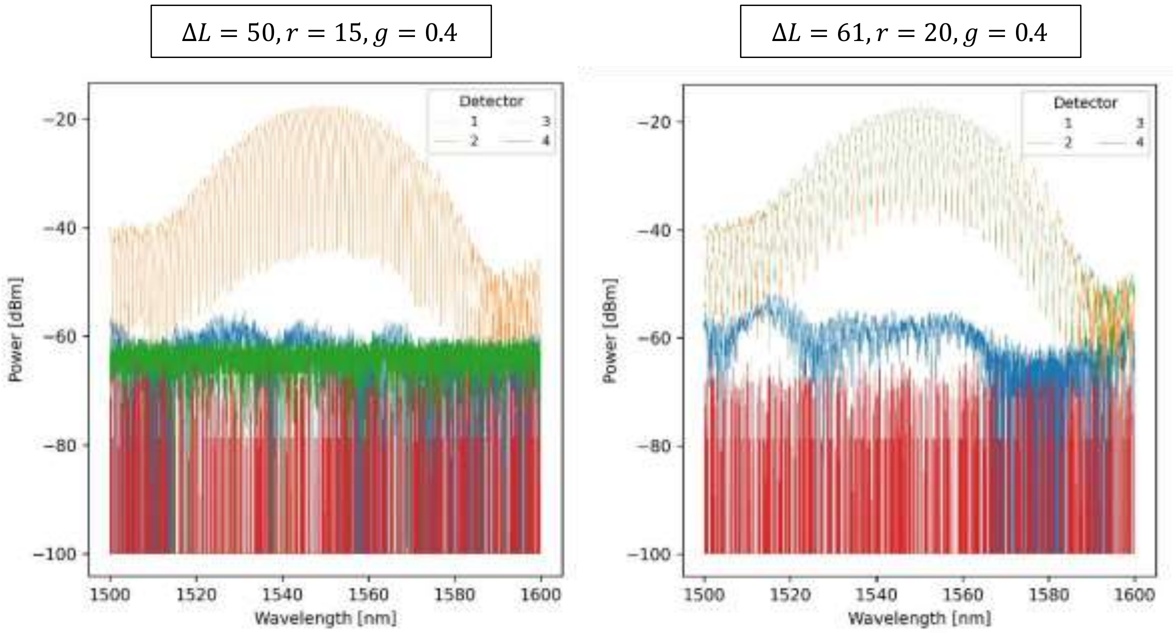


Fig. 8: Measured transmission spectra of MRR assisted MZIs

TABLE II

TE MRR assisted MZI Designs and FSRs at λ = 1550 nm (n_g(TE) = 4.2163)

Device	ΔL (μm)	Radius, r (μm)	Gap, g (μm)	FSR (nm)
Ring_mzi2	49.298	15	0.2	2.77
Ring_mzi3	40.918	20	0.2	1.37
Ring_mzi5	50.118	15	0.4	0.95
Ring_mzi6	61.238	20	0.4	2.81

VII. Analysis

A comparison table is shown bellow of simulated and measured FSR value of MRR assisted MZI.

TABLE III

Comparison of simulated and fabricated FSR at λ = 1550 nm

Device	ΔL (μm)	Radius, r (μm)	Gap, g (μm)	FSR (sim.) (nm)	FSR (fab.) (nm)	Deviation (%)
Ring_mzi2	49.298	15	0.2	11.34	2.77	75.57
Ring_mzi3	40.918	20	0.2	12.65	1.37	89.17
Ring_mzi5	50.118	15	0.4	6.39	0.95	85.13
Ring_mzi6	61.238	20	0.4	4.86	2.81	42.18

The fabricated FSR values are very lower than the simulated values with a deviation of 42-89%.

VIII. Conclusion

References

[1]. R. J. Bojko, J. Li, L. He, T. Baehr-Jones, M. Hochberg, and Y. Aida, "Electron beam lithography writing strategies for low loss, high confinement silicon optical waveguides," J. Vacuum Sci. Technol. B 29, 06F309 (2011)

[2]. Lukas Chrostowski, Michael Hochberg, chapter 12 in "Silicon Photonics Design: From Devices to Systems", Cambridge University Press, 2015

[3]. http://siepic.ubc.ca/probestation, using Python code developed by Michael Caverley.

[4]. Yun Wang, Xu Wang, Jonas Flueckiger, Han Yun, Wei Shi, Richard Bo-jko, Nicolas A. F. Jaeger, Lukas Chrostowski, "Focusing sub-wavelength grating couplers with low back reflections for rapid prototyping of silicon photonic circuits", Optics Express Vol. 22, Issue 17, pp. 20652-20662 (2014) doi: 10.1364/OE.22.020652

[5]. www.plcconnections.com, PLC Connections, Columbus OH, USA..

[6]. http://mapleleafphotonics.com, Maple Leaf Photonics, Seattle WA, USA